

Few-Mode Fibre for Noise Transfer Mitigation in Co-Raman Amplified Unrepeated Transmission

A. Busson⁽¹⁾, M. Quenu⁽¹⁾, J. Courty⁽¹⁾, B. Renot⁽¹⁾, H. Bissessur⁽¹⁾

⁽¹⁾ Alcatel Submarine Networks, France, alexis.busson@asn.com

Abstract We present a real-time Unrepeated experiment using a short section of Few-Mode Fibre at the transmitter side to mitigate non-linear degradation due to Co-Raman amplification. We demonstrate an improvement up to 0.4 dB of Q factor margin compared to classical fibre with large effective area. ©2026 The Author(s)

Introduction

Unrepeated optical fibre transmission systems operate without in-line active amplifiers, they are nevertheless achievable over hundreds of kilometres thanks to the Raman effect [1-4]. They usually take advantage of Raman amplification, both in Co and/or Counter propagated directions and in specific cases resort to Remote Optically Pumped Amplifier (ROPA).

In Co propagated Raman amplification, transmitted data and Raman pumps are propagated in the same direction from transmitter (TX) to receiver (RX) in order to increase the transmission distance. However, co-Raman amplification also generates lasing interferences which is visible as noise in the band of signal (grey line in Fig. 1), and results in a degradation of the Forward Error Correction (FEC) limit with Uncorrected Blocks (UB) appearing up to more than 1 dB above the theoretical FEC limit. Consequently, usage and benefit of co-Raman amplification become limited while Raman pump powers are expected to increase soon.

In this paper, we explore the potential to replace a portion of Pure Silica Core Fibre (PSCF) that is currently used in Unrepeated systems with a standard effective area (A_{eff}) of $80 \mu m^2$ with the same length of Few Mode Fibre (FMF) at the TX Submarine Line Terminal Equipment (SLTE). The use of bimodal optical fibre has already been evaluated through either numerical simulation for a 12 x 100 km repeated

transmission [5] or experimentally with a loop using N times 2 x 76 km with repeaters [6]. Concerning Unrepeated transmission, a quasi-single mode fibre [7] has been evaluated in a non-deployed link, i.e., using a variable attenuator to adjust the span loss, which is not fully representative of a multimode fibre in terms of noise transfer. In our work, we show that, in the presence of co-Raman amplification, propagating through a few kilometres of FMF at the transmitter end of an unrepeated optical line does not cause lasing interference (blue line in Fig. 1), unlike when classical PSCF with an A_{eff} of $80 \mu m^2$ is used (grey line in Fig. 1).

We report a comparison between the introduction of a short section of a step-index 2-LP-mode fibre with an A_{eff} of $219 \mu m^2$ for mode LP_{01} (2-LP-FMF), versus Single Mode Fibre (SMF) with a large A_{eff} of $150 \mu m^2$ (SMF- $150 \mu m^2$), and find that below 20 km the 2-LP-FMF outperforms the SMF- $150 \mu m^2$ when we transmit $9 \times 100 \text{ Gb/s}/\lambda$ carrying QPSK.

Finally, we report a fully loaded C-band transmission over 200 km with 30 channels carrying PCS-64QAM at $1 \text{ Tb/s}/\lambda$ and show that by adding only 1 km of 2-LP-FMF as the launch fibre at the transmitter side, all the channels are UB free after FEC decoding, which is not the case with only PSCF.

Experimental Set-up

The set-up of the experiment is shown in Fig. 2.a and involves 3 real-time modems operating with a FEC overhead of 16.7 % and adjustable with variable baud rates, modulations formats, entropies, and net data rates. The 3 transmitters are shaped with a Root Raised Cosine (RRC) filter with a roll-off factor of 0.1.

As we have only 3 real-time modems working with these variable parameters in our lab, we emulate a Dense Wavelength Division Multiplexing (DWDM) comb with the association of a broadband source using Amplified Spontaneous Emission (ASE) followed by a Wavelength Selective Switch (WSS). The ASE shaping is implemented such that the spectral shape of the actual data

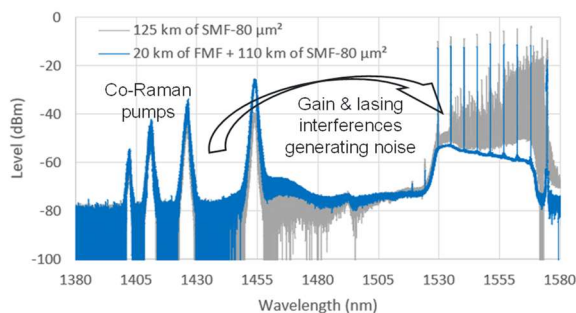


Fig. 1: Optical spectra of 4 co-Raman pumps from 1400 nm to 1455 nm and 8 DFB lasers in the C-band over 125 km of PSCF only (grey) and over 20 km of 2-LP-FMF followed by 110 km of PSCF (blue).

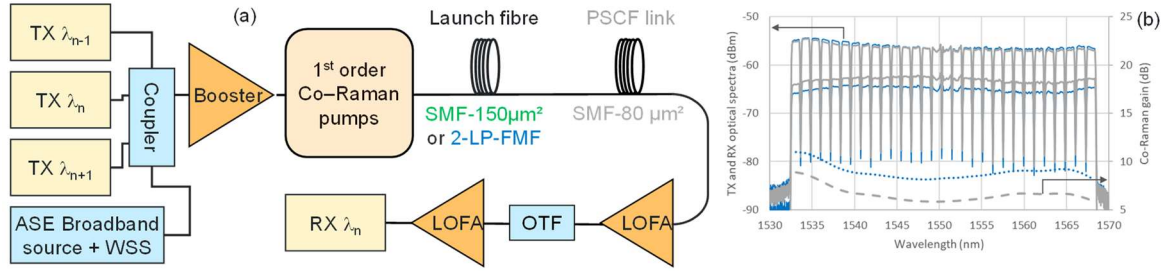


Fig. 2: Experimental set-up (a). Spectra at the transmitter and receiver for 200 km of PSCF (grey) and with 1 km of 2-LP-FMF added at the transmitter side (blue). The corresponding co-Raman gains for 30 channels are represented by the grey dashed and blue dotted lines respectively (b).

channel is replicated. Then an optical coupler combines the DWDM comb with the 3 real-time modems. The link primarily consists of PSCF having an A_{eff} of $80 \mu\text{m}^2$ and an attenuation around 0.17 dB/km at 1550 nm to which we add at the TX a section of either a SMF having a large A_{eff} of $150 \mu\text{m}^2$ or an FMF, with an attenuation of 0.15 dB/km and 0.18 dB/km respectively. Review and design of the FMF are described in [8-9] while the specific case of 2-LP-mode fibre is studied in [10]. The 3 transmitters, coupled with the DWDM loading comb, pass through a booster, then we take advantage of Co propagated Raman amplification at the TX SLTE. The co-Raman amplifier consists of four pumps from 1400 nm to 1455 nm with an adjustable power up to 660 mW for each pump.

At the receiver end, a Line Optical Fiber Amplifier (LOFA) is used to increase the sensitivity of the transponder, and the signal passes through an Optical Tunable Filter (OTF) to keep only the measured data channel.

The experiment consists of measuring the Bit Error Ratio (BER) of all the transmitted data channels one by one, by removing the corresponding wavelength λ_n as well as the two adjacent neighbours from the noise loading comb and tuning the real-time transmitters to channels λ_n , λ_{n-1} , and λ_{n+1} . Once the BER and the signal power of λ_n are measured, the wavelength is shifted to λ_{n+1} to measure the noise level at λ_n and calculate the Optical Signal to Noise Ratio (OSNR) at TX and RX; then the noise loading comb is configured for the next λ until the last channel is measured.

Results for different lengths of launch fibre

In a first step, we configure the set-up to transmit 9 channels carrying QPSK over 335 km with only PSCF, which represents a span loss of 58.6 dB ; the baud rate is 53.5 GBd shaped with RRC 0.1 , and the Channel Spacing (CS) is 59.38 GHz . For this configuration, we test different total co-Raman pump powers and find an optimum average FEC margin of 1.25 dB for 1.8 W . The FEC margin is defined as the difference between the

actual Q factor and the Q factor at FEC limit, which is measured in each configuration by decreasing the power of transmitters, keeping the co-Raman pumps constant. Here we define the FEC limit as 0 UB for 60 seconds , which represents an error ratio below $2\text{E-}12$ after decoding. The FEC limit is measured in the DWDM condition with the actual CS and channel power profile used to measure the Q factor performance for all the channels. Keeping the same 335 km PSCF link, we extend it by adding 5 km of SMF- $150 \mu\text{m}^2$ at the transmitter. For this configuration we find an optimal FEC margin of 1.1 dB with 2.2 W of total co-Raman pump power. In a third experiment, we replace the 5 km of SMF- $150 \mu\text{m}^2$ by 5 km of 2-LP-FMF and find the optimal performance for 2.64 W of total co-Raman pump power, which is the maximum we can get with our co-Raman pumping unit; we measure a FEC margin of 1.2 dB . The span loss for the 2 last configurations are 59.2 dB and 60 dB respectively.

We repeat the SMF- $150 \mu\text{m}^2$ versus 2-LP-FMF comparison for different lengths of launch fibre while keeping the total link length at 340 km . The comparison is shown in Fig. 3 and demonstrates that the 2-LP-FMF improves the FEC margin compared to the SMF- $150 \mu\text{m}^2$ for a launch section up-to 20 km while above that, the low losses of the SMF- $150 \mu\text{m}^2$ fibre becomes more

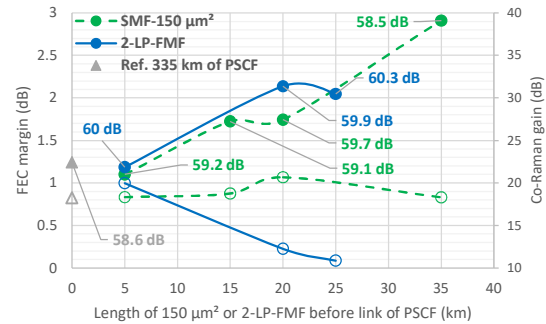


Fig. 3: Average over 9 channels of FEC margin (solid symbols) and corresponding co-Raman gain (empty symbols) for different lengths of launch fibre (total length = 340 km); triangle is the reference with 335 km of PSCF, dashed and solid lines are with SMF- $150 \mu\text{m}^2$ and 2-LP-FMF respectively. Numerical values represent the span loss in dB.

advantageous than the nonlinear penalty gain using 2-LP-FMF. For these two particular cases with 20 km of launch fibre in which the span losses are similar, the FEC margin with 2-LP-FMF outperforms the SMF-150 μm^2 by 0.4 dB. Stability measurements over several hours show similar variations in both cases.

Comparison between a link of PSCF and adding 1km of FMF at the transmitter

As a next step, we keep 9 channels carrying QPSK and compare 335 km of SMF-80 μm^2 versus the identical link to which we add only 1 km of 2-LP-FMF at the TX, representing a total link of 336 km. For these two configurations, the roll off factor and the CS remain identical at 0.1 and 59.38 GHz respectively. The result of the average of the 9 channels for each configuration is shown in Fig. 4 and demonstrates that adding 1 km of 2-LP-FMF after the TX SLTE, first we get an improvement of the FEC margin of 0.2 dB, and secondly, we have not yet reached the nonlinear threshold while the total co-Raman pump power is limited to 1.9 W with SMF-80 μm^2 only.

Following these results, we investigate a configuration with 30 channels carrying PSC-64QAM with an entropy of 4.86 bit/symbol over 200 km; the baud rate is at 129.8 GBd and all the channels transport 1 Tb/s/ λ with a CS of 150 GHz. The roll-off factor remains at 0.1. The optical pre-emphasis, and co-Raman pump powers are adjusted to keep a Q factor performance as flat as possible over the 30 transmitted data channels with the objective to maintain them UB free after FEC decoding. Fig. 5 shows the comparison between the link of 200 km with only PSCF (a) versus the same link to which we add 1 km of 2-LP-FMF as launch fibre at the transmitter side (b). TX and RX spectra as well as corresponding co-Raman gains are represented in Fig. 2.b. The span loss are 35.7 dB and 36.7 dB for cases (a) and (b) respectively. In case (a), the average FEC

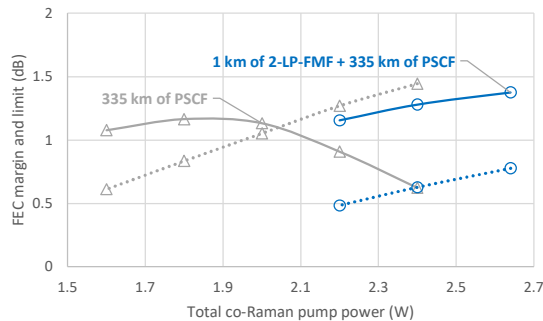


Fig. 5: Results for 9 channels carrying QPSK over 335 km. FEC margin (solid lines) and normalized FEC limit (dotted lines) as a function of total co-Raman pump power. Lines with triangle symbols represent the case with only PSCF and lines with dot symbols represent the case with 1 km of 2-LP-FMF added at the transmitter side. Results are the average of the 9 channels.

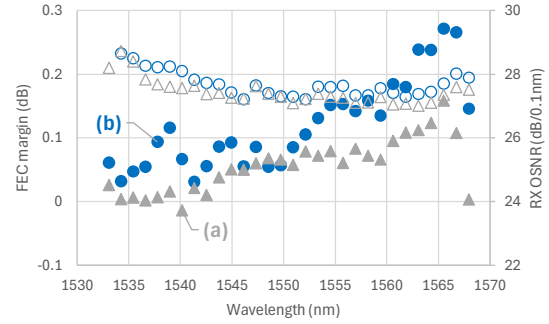


Fig. 4: Results for 30 channels carrying PCS-64QAM with: 200 km of PSCF only (a) and with 1 km of 2-LP-FMF added at the transmitter side (b); solid symbols represent Q factor margin and empty symbols the corresponding RX OSNR.

margin is 0.05 dB while the channel at 1540 nm presents UBs. The total co-Raman pump power is 1.05 W and gives an average co-Raman gain of 6.6 dB (Fig. 2.b), which is the optimum to balance the RX OSNR and the non-linear penalty for the first channels, i.e., increasing the pump power increases the number of UB appearances. The average TX and RX OSNRs are 35.1 dB/0.1 nm and 27.5 dB/0.1 nm respectively. When we add 1 km of 2-LP-FMF (b), we increase the average FEC margin to 0.12 dB with all channels UB free. Using 2-LP-FMF allows a total co-Raman pump power of 1.8 W, which gives 8.95 dB of co-Raman gain. The average TX and RX OSNRs are 34.8 dB/0.1 nm and 27.7 dB/0.1 nm respectively.

Conclusions

We report FEC margin measurements comparing an Unrepeated transmission over standard PSCF having an A_{eff} of 80 μm^2 combined at the transmitter side with either an SMF having a large A_{eff} of 150 μm^2 or a 2-LP-FMF. Transmitting 9 real-time channels carrying QPSK, we observe that 2-LP-FMF presents better performance than SMF-150 μm^2 below 20 km, while above 20 km the latter becomes more advantageous due to its low loss. The potential of the 2-LP-FMF as a launch fibre is confirmed with an Unrepeated DWDM transmission operating at 129.8 GBd in PCS-64QAM. Transmitting 30 x 1 Tb/s/ λ amplified with co-Raman pumping, we achieve a capacity of 30 Tb/s above 201 km using C-band technologies in a unique single core fibre and without resorting to ROPA or counter-Raman amplification.

These experiments show that the use of 2-LP-FMF as a launch fibre at the TX SLTE can be a promising candidate to mitigate non-linear penalty when co-Raman amplification is used in Unrepeated optical transmission without suffering from coupling between the fundamental LP_{01} mode that carries the transmitted data channels and the higher-order LP_{11} mode.

Acknowledgements

We would like to thank M. Bigot, P. Sillard, H. Maerten and M. O. Blais from Prysmian for the loan of the 2-LP-FMF as well as the information regarding it.

References

- [1] R. H. Stolen and E. P. Ippen, "Raman gain in glass optical waveguides," *Appl. Phys. Lett.*, vol. 22, pp. 276–278, 1973, DOI: [10.1063/1.1654637](https://doi.org/10.1063/1.1654637)
- [2] E. Desurvire, J. P. Pocholle, J. Raffy, and M. Papuchon, "Two Applications of Nonlinear Guided Optics" in *L'Optique Guidée Monomode*, Masson, ed. (1985).
- [3] E. Lefranc, "Les systèmes sans répéteur" in *Télécoms optiques*, Hermes, ed. (2003).
- [4] J. Bromage, "Raman Amplification for Fiber Communications Systems", in *Journal of Lightwave Technology*, vol. 22, n°1, pp. 79-93, 2004, DOI: [10.1109/JLT.2003.822828](https://doi.org/10.1109/JLT.2003.822828)
- [5] C. Koebele, M. Salsi, G. Charlet, and S. Bigo, "Nonlinear Effects in Long-Haul Transmission over Bimodal Optical Fibre", in *European Conference and Exhibition on Optical Communication*, Mo.2.C.6, 2010, DOI: [10.1109/ECOC.2010.5621347](https://doi.org/10.1109/ECOC.2010.5621347)
- [6] F. Yaman, N. Bai, Y. K. Huang, M. F. Huang, B. Zhu, T. Wang, and G. Li, "10 x 112Gb/s PDM-QPSK transmission over 5032 km in few-mode fibers", in *Optics Express*, vol. 18, issue 20, pp. 21342-21349, 2010, DOI: [10.1364/OE.18.021342](https://doi.org/10.1364/OE.18.021342)
- [7] Y.K. Huang, E. Ip, S. Zhang, F. Yaman, J. D. Downie, W. A. Wood, A. Zakharian, J. Hurley, S. Mishra, Y. Aono, E. Mateo, and Y. Inada, "20.7-Tb/s Repeater-less Transmission Over 401.1-km Using QSM Fiber and XPM Compensation via Transmitter-side DBP", in *Opto-Electronics and Communications Conference*, 2016
- [8] M. Bigot-Astruc, D. Boivin, and P. Sillard, "Design and Fabrication of Weakly-Coupled Few-Modes Fibers", in *IEEE Photonics Society Summer Topical Meeting Series*, TuC1.1, 2012, DOI: [10.1109/PHOSST.2012.6280766](https://doi.org/10.1109/PHOSST.2012.6280766)
- [9] P. Sillard and D. Molin, "A Review of Few-Mode Fibers for Space-Division Multiplexed Transmissions", in *European Conference and Exhibition on Optical Communication*, Mo.3.A.1, 2013, DOI: [10.1049/cp.2013.1272](https://doi.org/10.1049/cp.2013.1272)
- [10] D. Boivin, M. Bigot-Astruc, M. Travagnin, and P. Sillard, "Weakly-coupled Few-mode Fibers for Single-mode and Mode-division-multiplexed Transmissions", in *Optical Fiber Communication Conference and Exposition and the National Fiber Optic Engineers Conference*, OTh3K, 2013, DOI: [10.1364/OFC.2013.OTh3K.6](https://doi.org/10.1364/OFC.2013.OTh3K.6)